AzurESky "ensure quality of test"	**PERSONNEL COMPETENCE DECLARATION FORM**		
	Document no: ASMF6.1 FORM		
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**	
	Approved by:	**Nonhlanhla Msibi - Operations Manager**	

TECHNICAL SIGNATORY COMPETENCE DECLARATION FORM

PERSONNEL

Employee name: Maponya TMJ

ID Number: 0109196028087

Coy Number: 04

I declare that I am competent, trained, and authorized to perform my assigned tasks in accordance with the company's management system. I understand and agree to follow all applicable instructions as per the outlined procedure. I accept responsibility of being a technical signatory in accordance with SANAS R03.

Tasks:

Quick ash ☑

ISO Volatile matter ☑

ISO Inherent Moisture ☑

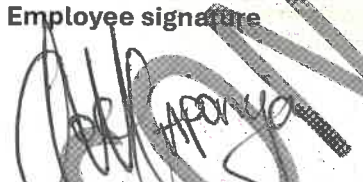
ISO Calorific Value ☑

Total Sulphur ☑

Sample Preparation ☑

Technical Signatory Assessment ☑

Employee signature



AUTHORIZING OFFICER

Name: N. P. Msibi

Designation: Operations Manager

Coy number: 01

CONTROLLED DOC

Effective date	Rev status	Next review date
07/2025	00	07/2028

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	Document no: ASMF6.1 FORM		
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**	
	Approved by:	**Nonhlanhla Msibi - Operations Manager**	

I declare that the above-named individual has been assessed and is authorized to perform the specified tasks in accordance with the company's management system and competency requirements.

Authorizing Signature



CONTROLLED DOC

Effective date	Rev status	Next review date
07/2025	00	07/2028

AzurESky (PTY) LTD
Unit 3, 17 Makriel Rd, Wadeville, Germiston, 1421, Gauteng, South Africa
Tel: +27 (0) 13 690 2867/2818 Fax: +27 (0) 13 690 2818

To	AZURESKY	Reference number:	SEPTEMBER 2025 1610
For Attention	NONHLANHLA MSIBI	Sample submitted:	2025/09/09
Tel:		Sample description:	
Fax:		Contact Person:	MAPONYA
Cell:	082 786 8149	Report date:	2025/11/18
Address \ Email:	NONHLANHLA@AZURESKY.IT.COM	Report status:	Final
		Number of pages	1 OF 2

TEST REPORT

			Results reported on an Air-Dry Basis							
Lab.no.	Sample name	Description	Inherent Moisture	Ash Content	Volatile Matter	Fixed carbon	Calorific Value	Total sulphur	Phos *	Total H2O
			%	%	%	Calc / %	MJ/Kg	%	%	%
0	SEPT 1 - TSHEPO	SEPT 1 - TSHEPO	1.8	21.1	24.1	53.0	25.31			

Tests marked with an asterisk (*) does not form part of the scope of SANAS accredited tests. All other tests are not included in the schedule of accreditation for this laboratory.

(Condition of test item if abnormal / additional information)

Technical Signiture

A signed copy of this report is available at the laboratory

Results reported are only representative of the sample received for testing

While every endeavour has been made to ensure that the results reported are accurate, the laboratory shall not be liable for any error nor for any damage or loss what so ever that may arise from the issuing of this certificate. This certificate may not be reproduced in part or full without written permission from the Technical Managers of the laboratory.

Refer to Addendum to this report for method identification as well as scope of accreditation

AzurESky	AZURESKY PTY LTD	Document # AZCA-006-2	Revision# 0
Subject	TECHNICAL SIGNATORY		

TECHNICAL SIGNATORY TRAINING AND ASSESSMENT

Name and Surname: Thepo Maponya

Date: 08/11/2025p

Total = 50 marks Questions = 30

45/50 100%

SECTION A: Roles, Responsibilities and SANAS Requirements (R03-10 & R04) (12 marks)

1. Explain the primary responsibilities of a Technical Signatory as outlined in R03-10. How does this role ensure the technical validity of test results? (2)

Answer:

* Technical Signatories' primary responsibility is to approve test report.
* TS ensures the technical validity of test results by a show of their signature on test reports.

2. Describe the process your laboratory follows to nominate and approve a new Technical Signatory. Which documentation and evidence must be submitted to SANAS? (2)

Answer:

* Be conversant with the 7 Sanas documents
* Be conversant with all the methods in the CABs scope of accreditation
* Be appointed by the CAB as an applicant technical signatory to SANAS
* Pass theoretical assessment set up by the CAB for all methods

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3. Discuss three qualification or experience requirements that must be met before approval as a Technical Signatory under R03-10. (2)

Answer:

* 3 years experience within the relevant field
* Lab must appoint a new TS through an internal approval process that includes competency evaluation, management endorsement, technical Signatory assessment.
*

4. Explain the difference between a Nominated Representative and a Technical Signatory according to R03-10, and why these roles must be clearly defined. (2)

Answer:

A NR is the person of contact between the CAB and SANAS and a technical signatory authorised approves technical results

5. Describe how the Technical Signatory ensures correct use of the SANAS logo and reference to accreditation in reports, per R04. Give one example of incorrect usage. (2)

Answer:

* A technical signatory ensures that the SANAS logo is to be used in technical reports where over 50% of the methods included in the test report are accredited.

* when the SANAS logo is predominately larger in size than the company logo.

6. Outline the procedure for referencing SANAS accreditation on laboratory reports to maintain traceability. (2)

Answer:
Reports must correctly reference the accreditation by the inclusion of the SANAS logo, the CABs accreditation number and make reference to the accredited service (testing, inspection, calibration service)

SECTION B: Suspension, Withdrawal and Reinstatement (R51-10) (8 marks)

7. Under what circumstances may SANAS suspend or reduce a laboratory's scope? Provide an example. (2)

Answer:
If a CAB fails to meet accreditation requirements, this may be because there's Key staff member/personnel last and no evidence of their competencies, non-conformances not resolved after assessment.

8. State the laboratory's obligations during suspension or withdrawal, covering both administrative and technical aspects. (2)

Answer:
* The CABs must discontinue the use of the SANAS accreditation logo and making any reference to accreditation in their reports
* Clients must be informed of the suspension, withdrawal.

9. Describe the process and evidence required for reinstatement following suspension. (2)

Answer: The lab must submit corrective action and evidence of compliance of affected areas.

10. Discuss the possible impacts of accreditation withdrawal on laboratory credibility and client confidence. (2)

Answer:
It may damage reputation of the CABs credibility, questions the reported data, contract breakdowns and damage the confidence of the CAB

SECTION C: Proficiency Testing and Interlaboratory Comparison (R80-06) (8 marks)

11. Explain the importance of proficiency testing (PT) participation for accredited laboratories and how it supports confidence in results. (2)

Answer:
* PT reveals that a CAB can produce accurate and comparable results against other competent labs. It gives the CAB reliability and confidence in the results they provide their clients with.

12. List criteria to consider when selecting a suitable PT or interlaboratory comparison provider. (2)

Answer:
* The PT provider must show competence against the relevant standard (17043)
* Assess the required frequency

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13. How should a Technical Signatory respond to unsatisfactory PT performance? Describe the steps for investigation and correction. (2)

Answer:
The TS must open a non-compormance to investigate the root cause. Find ways to assess and close the root cause (possible calibration, ~~re~~equipment checks etc) and document the outcome of the investigation.

14. Discuss how PT data can be used for ongoing staff competence verification. (2)

Answer:
Previous PT data has a known value from the PT report. The CAB can use ~~the~~ the previous PT samples to assess staff competence and gauge them by how far of their result are compared to the known value using reproducibility limits.

SECTION D: Calibration and Intermediate Checks (TR25) (10 marks)

15. Define "intermediate check" per TR25 and explain its purpose in maintaining measurement confidence. (2)

Answer:
This is a performance check in the form of equipment verification that provide accuracy of equipment and how stable it is during the validity period. This gives the CAB confidence in their measurement instruments.

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16. Give three examples of coal-testing instruments requiring intermediate checks and describe what parameter each check verifies. (2)

Answer:
Balance verification - Proximates, Physical Tests (PSD, HGI, FTS)
CV, TS
Oven Verification - Total moisture, AI, Inherent moisture
Furnace Verification - Volatile, Ash (Quick & ISO)
Knvironmental control and monitoring equipment
: These affects CV, & Inherent moisture, volatile mostly

17. Explain the difference between calibration and verification, giving one laboratory example. (2)

Answer:
Calibration is setting an instrument's performance detection & program against known standards.
Verification is an intermediate check that assesses if an instrument is still accurate and stable during its validation period.
E.g. Balance is calibrated by an accredited facility and the lab performs balance verification prior to use.

18. Describe how the Technical Signatory ensures calibration certificates meet traceability requirements in TR25. (2)

Answer:
* The calibration certificate shall possess measurements that are traceable to an International SI units (kg, °C,)
* The CAB must use a facility that is SANAS accredited to ensure an unbroken chain of traceability.

19. Discuss the impact of using equipment that has missed scheduled checks on test validity and accreditation compliance. (2)

Answer:

* This can negatively affect the confidence in results that would be produced by the said equipment
* The validity of such results can be questioned.
* It is an non automatic non-conformance as it breaches compliance against ISO 17025, TR25

SECTION E: Method Validation and Verification (TR26) (8 marks)

20. Differentiate between validation and verification of analytical methods as per TR26 and indicate when each applies. (2)

Answer:

* Validation is used when there's an introduction of a new method entirely or when there's a new instrument inserted into an already established method. This assesses the technical capability of a said method using performance characteristics.
* Verification uses known standards or CRM to verify employed analytical methods.

21. Identify three performance characteristics evaluated during method validation under TR26. (2)

Answer:

Accuracy, precision, bias, Robustness

22. Explain how the Technical Signatory ensures validated methods remain fit for purpose, citing two review mechanisms. (2)

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Answer:
* The inclusion of the said method into a competent PT scheme to assess ~~valid~~ if it produces valid results against other competent labs
* The review of control charts assess ~~the~~ if the method is still fit for purpose as it uses a known value as a basis

23. Describe the link between uncertainty of measurement and method validation, and its importance in coal testing decisions. (2)

Answer:
* For a method to be analytically valid, its UoM has to be assessed to address the level of doubt on a can place on the method.
* It gives comparability between labs and therefore influences decisions in terms of compliance.

SECTION F: Technical Competence and Laboratory Operations (4 marks)

24. Explain how the Technical Signatory ensures accuracy and repeatability in: Ash Determination, Calorific Value, Inherent Moisture, ISO Volatile Matter, and Sample Preparation. (2)

Answer:
* For the general analysis, accuracy is ensured by the use of a CRM during analysis of samples.
* Sample repeatability is ensured against established limits derived from ISO standards, replicate testing
* Use of validated methods and calibrated equipment ensures accuracy of test results.

25. Reflect on how SANAS documents R03-10 to TR26 collectively promote data integrity and continual improvement in an accredited coal laboratory. (2)

Answer:
R03-10 defines key competence roles, R04 outlines correct usage of TR25 ensures the use of calibrated & verified equipment
TR26 incorporates validation of analytical methods developed.

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SECTION G: Applied Analytical Calculations (Laboratory Data Exercise) (10 marks total – 5 questions × 2 marks)

26. Convert the Volatile Matter result from air-dry to dry-basis. (2)

Answer: $DB = \left(\frac{100}{100 - IM}\right) \times AD$

$= \left(\frac{100}{100 - 3,83}\right) \times 27,62$

$= 28,72$ 2

27. Calculate the Total Moisture %. (2)

Answer: $TM = \left(\frac{100 - SM}{100}\right) \times RM + SM$

$= \left(\frac{100 - 4,26}{100}\right) \times 3,01 + 4,26$

$= 2,88 + 4,26$

$= 7,14\ \%$ 2

28. Convert the % Ash result from air-dry to as-received basis. (2)

Answer: $Ash = \frac{Ash(Ad) \times (100 - TM)}{100}$

$= 20,38 \times 0,9273$

$= 18,89$ 2

29. Convert the Calorific Value from MJ/kg to kcal/kg. (2)

$CV\ (KCAL/kg)^{TMS} = (CV\ (MJ/kg) \times 238,92$

$= 25,52 \times 238,92$

$= 6097,24\ kcal/kg$ 2

Answer:

30. Apply the appropriate correction for Volatile Matter and explain its purpose. (2)

Answer:
By definition Volatile matter has to be determined in the absence of ~~air/oxygen~~ moisture/air meaning what is determined must be free from moisture hence the correction for moisture.

AzurESky	AZURESKY PTY LTD	Document # AZCA-005	Revision# 0
Subject	ASSESSMENT FOR SAMPLE PREPARATION		

PERSONAL DETAILS:

Name:	Maponya TMJ
Job Title:	Laboratory Supervisor

ASSESSMENT:

Attempt:	01
Date:	06/11/2025
Time:	10:00

ASSESSORS:

Name	1.- N Mbibi	
	2.	

AzurESky	AZURESKY PTY LTD	Document # AZCA-005	Revision# 0
Subject	ASSESSMENT FOR SAMPLE PREPARATION		

Instructions and Information

1.	The competency and development tool is based on the Sample Preparation **Desired Outcome:** At the end of this module the employee should be able to perform the determination of ash content in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the Sample Preparation. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **20** points of which **2** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzureSky	AZURESKY PTY LTD	Document # AZCA-005	Revision# 0
Subject	ASSESSMENT FOR SAMPLE PREPARATION		

Assessment Sample Preparation

Date: 06/11

Candidate name: Ishepo **Staff No:** 04

Candidate Signature: Ishepo

No	Question:	Points	Score
1.	Name two apparatus used to mix sample for preparation Splitter & crusher, homogenizing drum	2	Splitter (Rusher
2.	What is the purpose of crushing the sample? Reduction of sample nominal topsize	2	
3.	Why do we air dry the sample if it is visible wet? This is so the sample can be at equilibrium with the lab environment. To avoid blockages on equipment	2	
4.	Why is housekeeping such an important part of Good Laboratory Practices (GLP)? This is to avoid any possible cross contamination amongst sample.	1	
5.	Name 5 steps to be followed air-drying the sample ① Receive & weigh sample ② On a clean floor, spread sample evenly ③ Dry sample until it changes from a darker colour to a light greyish colour ④ If conditions don't permit, air dry sample at 40°C not more than	* 5	
6.	At what temperature is the test sample dry for the general analysis? 40°C	* 1	
7.	When is the pulveriser verification to be done? Pulverizer verification to be done prior to use.	1	
8.	What mass is extracted from the homogeneity drum during packaging? 600g x 10 at random	1	
9.	What are the acceptance criteria for crusher efficiency? efficiency must be greater or equal to 90%	1	

10.	Samples should be stored into a non-labeled bottle (True or false) Support your answer. False, non-identified sample will bring about confusion and ~~t~~errors	2	
11.	Reference samples are kept in the lab for future disputes by the clients. (True or false) True	1	
12.	How frequently must stored samples be checked for stability? Once per month	1	
	Total:	20	20

Comments by Assessor(s): Competent

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent** **and** *may / may not* proceed* **to the Assessment Panel for authorisation.**

* Delete as applicable.

Assessor Name N. Mabi **Signature:** _____ **Date:** 6/11/2025

If you do not agree with the feedback, please discuss this. If you can not come to an agreement, ask for an explanation of the appeals procedure.

I declare that I am satisfied with this assessment and can perform these tasks.

Learner Signature: _____ Date: 06/11

Senior analyst Signature: _____ Date: _____

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-004	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF THE ASH CONTENT (COAL-QUICK ASH METHOD)		

PERSONAL DETAILS:

Name:	Maponya TMJ
Job Title:	Laboratory Supervisor

ASSESSMENT:

Attempt:	01
Date:	06/11/2025
Time:	10:00

ASSESSORS:

Name	1. N. Moibi	
	2. EB Nyambi	

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-004	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF THE ASH CONTENT (COAL-QUICK ASH METHOD)		

Instructions and Information

1.	The competency and development tool is based on the ISO 1171 **Desired Outcome:** At the end of this module the employee should be able to perform the determination of ash content in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with the Ash Content determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the Ash Content. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **23** points of which **13** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.